Brief explanation of Confusion Matrix:

- **Accuracy**: The proportion of total predictions the model got right (both positives and negatives).
- **Precision**: Out of all cases the model said were positive, how many were actually positive.
- **Recall**: Out of all actual positive cases, how many the model correctly identified.
- **F1 Score**: A balanced score that combines precision and recall, especially useful when data is imbalanced.

What is Balance and Imbalance data?

- **Balanced data** means the classes (e.g., positive vs. negative) have roughly equal representation, while **imbalanced data** means one class significantly outnumbers the other—common in fraud detection, disease diagnosis, etc.

Which score is important for the model evaluation - how do we consider them (Very brief answer)

- The importance depends on the problem: **accuracy** is good for balanced data, while **precision**, **recall**, and **F1 score** matter more for **imbalanced data** or when **false positives/negatives** have different costs.

Why should we worry about FP/FN cost ?

- We should worry about **False Positives (FP)** and **False Negatives (FN)** because they have **real-world consequences**—e.g., a **FN in cancer detection** means missing a sick patient, while a **FP in fraud detection** may wrongly block a legitimate transaction. The cost depends on the **application's impact**.

🏥 Scenario: COVID-19 Test

The model predicts whether a person **has COVID-19** or **not**.
We test **100 people** (ground truth known), and get predictions as follows:

📊 Confusion Matrix Table

	Predicted: Positive	**Predicted: Negative**
Actual: Positive	**True Positive (TP)** = 40	**False Negative (FN)** = 10
Actual: Negative	**False Positive (FP)** = 5	**True Negative (TN)** = 45

💡 Breakdown

- ✅ **True Positive (TP = 40)**: Model correctly predicted 40 people *have* COVID.
- ❌ **False Negative (FN = 10)**: Model missed 10 people who actually *had* COVID (dangerous).
- ❌ **False Positive (FP = 5)**: Model wrongly said 5 people *have* COVID, but they didn't (unnecessary stress).
- ✅ **True Negative (TN = 45)**: Model correctly predicted 45 people *don't* have COVID.

📈 Total Evaluations: 100 people

- Actual Positives: 50
- Actual Negatives: 50
- Model Accuracy: (TP + TN) / Total = (40 + 45) / 100 = **85%**